Bryan Perry

MLIS 7310

13 November 2024

Metadata Application Profile

“Imperial Beach: A Thousand Days of Sanitation Crisis”

I: Introduction

I.i. Background

Since December 2021, my local city, Imperial Beach (located at the southwestern end of San Diego County, California), has been affected by millions of gallons of sewage released daily from the Tijuana River, originating from the neighboring city of Tijuana, Mexico (Kitt, 2024). This continuous sewage spill has been caused by the failure of the internationally owned International Boundary Wastewater Treatment Plant. The treatment plant has been undermaintained, and after being unable to keep up with additional burdens of a rapidly growing Tijuana, has finally failed, leading to untreated sewage entering the Pacific Ocean, leading to over 1000 days of continuous beach closures in Imperial Beach, California.

In additional the effects of water that is unsafe to swim in, Imperial Beach has been affected by overwhelming odors, local clinics have reported increases in patients complaining of respiratory and intestinal distress, and the Center for Disease Control recently conducted a survey of local residents to determine self-reported health ailments (Deng, 2024). The Governor of California has yet to declare a state of emergency, and neither has the federal government, and thus Imperial Beach mayor Paloma Aguirre has taken on additional burdens of advocating for her residents; far beyond what is normally expected of a small-city mayor (Yutig, 2024). As a local resident of Imperial Beach who regularly takes walks down to the

shoreline, the empty beach and the contaminated water signs have become perennial fixtures in my city.

I.ii Purpose

The purpose of this Metadata Application Profile (MAP) is to document and maintain provenance of photographs taken from local residents' perspectives, along with digitized corresponding articles from the local newspaper, *Imperial Beach and South County Eagle & Times.* The goal is to provide viewers with first-hand resident accounts of the ongoing water contamination crisis and provide a framework for cataloging or archiving these items for potential future use, such as in digital archives, or for research projects. The intended audience for this collection would be students and researchers studying international wastewater management, along with the effect of wastewater on small communities and government responses to these issues. This collection may also be helpful for those preparing citizen accounts and testimonies as evidence for regulatory bodies.

I.iii Scope

The scope of this collection includes photographs taken in Imperial Beach, California, during the ongoing wastewater contamination crisis (late 2021 to present/ late 2024), along with articles focusing directly on this crisis, from within the same timeframe. The collection can expand to include born-digital articles, but for the time being, media consists of digitized formats of printed articles from the *Eagle & Times* newspaper. Since the photographs have yet to be published and made publicly available, maintaining as much provenance and evidence of their authenticity is of paramount importance to this metadata application profile.

II: Metadata Application Profile

II.i. Elements

Element Name	DC Term	Obligation	Vocabulary	Example(s)
Name of Photo/ Title of Article	Title	Required		IMG_5632 “Circus Arrives on New Year’s Day”
Author	Creator	Required (Use “Anonymous” if none given)	LC Authorities (when applicable)	Gladwell, Malcolm, 1963-
Photographer	Creator	Required for photographs	LC Authorities (when applicable)	Arbus, Diane, 1923-1971
Topic	Subject	Required	Library of Congress Subject Headings	Water contamination; Beaches--California
Content of the item	Description	Required (provide summary)		Photograph of contaminated water notice sign
Date of Publication	Date	Required if available	ISO 8061	2024-10-24
Time and Date of Capture (Exif)	Date	Required if available	ISO 8061	1995-11-20T23:45:32
Date of Newspaper issue	Issued	Required for Newspaper Articles		2024/10/31
Medium of Item	Type	Required	DCMI Type	Image; video
Original source	Source	Required if available		New York Times; Wall Street Journal

Source publisher	Publisher	Required for journal/ newspaper articles		American Libraries Association; American Heart Association
Date coverage of newspaper	Coverage	Required if available	ISO 8061	2024-10-24/2024-10/31
Coordinates (Longitude and Latitude) of photograph	Spatial	Required if available	DCMI Point Encoding Scheme	north=38.556; east= 35.667
City/Region/ County	Spatial	Required	Thesaurus of Geographic Names	Los Angeles (County); Tōhoku-chihō (region (geographic))
State/Province	Spatial	Required	TGN	California (State); Alberta (Province)
Digital File Format	Format	Required	IANA Media Types	Text/pdf; video/ H264
Holder of the original photograph	Provenance	Required if available (provide either the owner, or holding institution)		Smith, John ; Anytown University Special Collections and Archives
Parent Newspaper Issue	IsPartOf	Required if available		Vol. 55, No. 24
Article that is a scan of a physical copy	IsFormatOf	Required if available (only if different from original)		Original printed magazine; VHS cassette tape
Language(s) in photograph or text	Language	Required if any text is displayed	ISO 639-2	eng; spa; jpn

Camera Type Used	Source	Required if available (provide model, no controlled vocab necessary)		• iPhone 12 • Samsung Galaxy S23 Ultra • Sony NEX-3n

II.ii Creation Process

This Metadata Application Profile uses Qualified Dublin Core as its standardized element set, with a number of elements interpreted from a different metadata standard. I chose Qualified Dublin Core for several reasons. Firstly, I wanted this metadata standard to be easily readable for students, as well as researchers. Secondly, Dublin Core's simplicity and non-mandatory but repeatable elements allows me to add and subtract elements based on the item in question. Digital photographs and digitized newspaper articles, while both similar in medium, are fundamentally different in how they are described, and using a metadata standard that allows me to include or omit elements and fields based on each item allows for a well-optimized MAP. Metadata elements incorporate and crosswalk a selection of fields from Exchangeable image file format (Exif), the–metadata standard used in digital photography–including resolution, file type, camera type, coordinates. The goal for mapping out image metadata is to preserve a reasonable level of provenance that lends credence to the authenticity of these photographs.

For this MAP, I would be creating metadata for both digital photographs, and digitized articles from printed newspapers. Starting with digital photographs, I determined that the necessary metadata fields should preserve a degree of provenance and information and demonstrate their authenticity. I first searched *Flickr.com* for

examples on how creators used metadata to provide authenticity for their own photographs. I noticed that images were described in Exchangeable image file format (Exif) structure, which is the metadata standard used in digital photography. As I took these photographs with an iPhone 12 Pro, I had access to the metadata in Exif form. Therefore, the next step was to figure out how to map Exif metadata into Dublin Core fields. After consulting Dr. Choi on determining equivalent Exif and DC fields, I consulted the DCMI website on how to format this information. Using my iPhone's metadata, I determined the values most important to map included file name, geocoordinates, time and date of capture, camera type, and file format. Referring back to Dr. Choi's suggestions, I chose the closest approximate DC element fields, taking advantage of the repeatability of these fields.

Using the appropriate vocabulary and formats for a number of element fields was the next challenge. For the date and time field, I noted that Exif uses the ISO 8061 format, beginning from the largest denomination of time, year, and ending with the smallest: seconds (or fractions of seconds). I then consulted the Dublin Core website, noted that DC also uses ISO 8061, and decided to map this field as this format standard. Next, I deferred to DC's Point Encoding Scheme as my guide for recording the longitude and latitude of photographs taken, and DCMI Type for the format of items. Since I wanted to also include the type of digital file format for my photographs, I searched DC's Format page, which prompted me to choose Internet Assigned Numbers Authority Media Types as my controlled vocabulary for this field. Lastly, since I am the original photographer of these pictures, adding a "Provenance" field that specifies

myself as the original custodian, information I deem important for verifying their authenticity.

Deciding on which elements were necessary for digitized articles was the next step for creating my metadata application profile. As the articles are part of a newspaper with regular weekly circulation, having fields that both specified the publication date, as well as the circulation dates, were necessary. To specify the parent volume and edition of the newspaper, I used the “IsPartOf” field, and the “Coverage” field to specify the time period of the newspaper issue, “Date” to refer to the specific date the newspaper was issued to the public; and “Publisher” to refer to the company that owns the newspaper. Since *Eagle & Times* does not always specify the name of the editor-in-chief, or the author of headline articles, my MAP allows users to use “Anonymous” in place of an unnamed writer. Finally, as the items being described focus on a specific geographic region, and on a particular subject matter, adding Spatial and Topic fields, respectively, were crucial to making these items more readily identifiable to audiences. Following guidelines from Dublin Core, I chose the Getty Thesaurus of Geographic Names (TGN) as my spatial controlled vocabulary, and the Library of Congress Subject Headings as my topical controlled subject vocabulary.

III Examples

Example 1:

Name of Photo	Title	IMG_5264
Photographer	Creator	Perry, Bryan
Topic	Subject	Beach Closures--California

Topic	Subject	Sewage disposal in the ocean--California.
Topic	Subject	Sewage disposal--Law and legislation--California--Tijuana River Estuary.
Topic	Subject	Water contamination
Content of the Item	Description	A photograph of a closed beach sign, with a pier in the background: Imperial Beach, CA.
Time and Date of Capture	Date	2024-11-04T16:37
Medium of Item	Type	Image
Coordinates	Spatial	north=32°34'47.5" ; west=117°07'58.8"
City/Region/County	Spatial	Imperial Beach (Inhabited Place)
City/Region/County	Spatial	San Diego (County)
City/Region/County	Spatial	Tijuana (Inhabited Place)
State/Province	Spatial	California (State)
State/Province	Spatial	Baja California (State)
Digital File Format	Format	Image/heic
Holder of the original photograph	Provenance	Photograph held by original creator, B. Perry
Language(s) in Photograph or text	Language	eng
Camera Type Used	Source	Apple iPhone 12 Pro

Example 2:

Name of Photo	Title	IMG_5227
Photographer	Creator	Perry, Bryan
Topic	Subject	Beach Closures--California

Topic	Subject	Sewage disposal in the ocean--California.
Topic	Subject	Sewage disposal--Law and legislation--California--Tijuana River Estuary.
Topic	Subject	Water contamination
Content of the Item	Description	A photograph of a closed beach sign due to contaminated water. Spanish language. Imperial Beach, CA.
Time and Date of Capture	Date	2024-11-03T06:13:31
Medium of Item	Type	Image
Coordinates	Spatial	north=32°34'44.9" ; west=117°07'57.8"
City/Region/County	Spatial	Imperial Beach (Inhabited Place)
City/Region/County	Spatial	San Diego (County)
State/Province	Spatial	California (State)
Digital File Format	Format	Image/heic
Holder of the original photograph	Provenance	Photograph held by original creator, B. Perry
Language(s) in Photograph or text	Language	spa
Camera Type Used	Source	Apple iPhone 12 Pro

Example 3:

Name of Photo	Title	IMG_0507
Photographer	Creator	Perry, Bryan
Topic	Subject	Beach Closures--California
Topic	Subject	Sewage disposal in the ocean--California.

Topic	Subject	Sewage disposal--Law and legislation--California--Tijuana River Estuary.
Topic	Subject	Water contamination
Content of the Item	Description	A handful of people enter the water on a June afternoon, despite contaminated water warning signs in the foreground. Imperial Beach, CA.
Time and Date of Capture	Date	2023-06-07T17:25:16
Medium of Item	Type	Image/heic
Coordinates	Spatial	north=32°34'50.1" ; west=117°07'57.6"
City/Region/County	Spatial	Imperial Beach (Inhabited Place)
City/Region/County	Spatial	San Diego (County)
State/Province	Spatial	California (State)
Digital File Format	Format	heic
Holder of the original photograph	Provenance	Photograph held by original creator, B. Perry
Language(s) in Photograph or text	Language	eng
Camera Type Used	Source	Apple iPhone 12 Pro

Example 4

Title of Article	Title	California Coastal Commission Votes to Support State of Emergency Declaration

Author	Creator	Anonymous
Topic	Subject	Beach Closures--California
Topic	Subject	Sewage disposal in the ocean--California.
Topic	Subject	Sewage disposal--Law and legislation--California--Tijuana River Estuary.
Topic	Subject	Water contamination
Content of the Item	Description	Newspaper article detailing the California Coastal Commission's unanimous vote to send a letter to President Biden, requesting a federal state of emergency over sewage contamination from the Tijuana River.
Date of Newspaper Issue	Issued	2024-10-17
Medium of Item	Type	Text/pdf
Original Source	Source	Imperial Beach and South County Eagle & Times
Newspaper Publisher	Publisher	Eckenroth Publications, Inc., dba Eagle Newspapers
Date of Newspaper Coverage	Coverage	2024-10-17/2024-10-23
City/Region/County	Spatial	Imperial Beach (Inhabited Place)
City/Region/County	Spatial	San Diego (County)
State/Province	Spatial	California (State)
Digital File Format	Format	pdf
Parent Newspaper Issue	IsPartOf	Vol. 40, No. 42
Article that is a scan of a physical copy	IsFormatOf	Original print newspaper issue.

Language(s) in Photograph or text	Language	eng

Example 5

Title of Article	Title	The Mayor’s Corner: Coastal Commission Joins the Fight for A State of Emergency on the Tijuana River Crisis
Author	Creator	Aguirre, Paloma
Topic	Subject	Beach Closures--California
Topic	Subject	Sewage disposal in the ocean--California.
Topic	Subject	Sewage disposal--Law and legislation--California--Tijuana River Estuary.
Topic	Subject	Water contamination
Content of the Item	Description	In a direct address to residents, Imperial Beach Mayor Paloma Aguirre provides an update on actions currently being taken to request a federal state of emergency on the ongoing sewage spill crisis from the Tijuana River.
Date of Newspaper Issue	Issued	2024-10-24
Medium of Item	Type	Text/pdf

Original Source	Source	Imperial Beach and South County Eagle & Times
Newspaper Publisher	Publisher	Eckenroth Publications, Inc., dba Eagle Newspapers
Date of Newspaper Coverage	Coverage	2024-10-24/2024-10-30
City/Region/County	Spatial	Imperial Beach (Inhabited Place)
City/Region/County	Spatial	San Diego (County)
State/Province	Spatial	California (State)
Digital File Format	Format	pdf
Parent Newspaper Issue	IsPartOf	Vol. 40, No. 43
Article that is a scan of a physical copy	IsFormatOf	Original print newspaper issue.
Language(s) in Photograph or text	Language	eng

IV References

Deng, J. (2024, October 4). *The tijuana river smells so bad, the CDC is coming to investigate*. Los Angeles Times. https://www.latimes.com/environment/story/2024-10-04/tijuana-river-cdc-survey-smell-illnesses

The Getty Research Institute. (2017, March 17). *Getty Thesaurus of Geographic Names® Online*. Getty Thesaurus of geographic names (Getty Research Institute). https://www.getty.edu/research/tools/vocabularies/tgn/?find=San%2BDiego%2Bcounty&place=&nation=&english=Y

Kitt, M. (2024, September 26). *After lengthy closure, Imperial Beach shoreline is (mostly) open*. Coronado Times. https://coronadotimes.com/news/2024/09/24/after-lengthy-closure-imperial-beach-shoreline-is-mostly-open/

Library of Congress. (2017, December 21). *Codes for the representation of names of languages*. ISO 639-2 Language Code List - Codes for the representation of names of languages (Library of Congress). https://www.loc.gov/standards/iso639-2/php/code_list.php

Library of Congress. (n.d.). *Library of Congress authorities (search for name, subject, title and name/title)*. Library of Congress . https://authorities.loc.gov/

Miller, S. J. (n.d.). *Examples of Metadata Functionality, Application Profiles, and Records*. ALA Store. https://alastore.ala.org/sites/default/files/MDC_Functionality-ApplicationProfiles-Records.pdf

Melnikov, A., Miller, D., & Kucherawy, M. (2024, November 5). *Media types*. Internet Assigned Numbers Authority. https://www.iana.org/assignments/media-types/media-types.xhtml#text

Rühle, S., Baker, T., & Johnston, P. (2017, September 1). *Creating metadata*. DCMI. https://www.dublincore.org/resources/userguide/creating_metadata/

Yutig, J. (2024, September 4). *Imperial Beach Mayor says community feels "abandoned" by California governor*. CBS San Diego. https://www.cbs8.com/article/news/local/cross-border-contamination/imperial-beach-mayor-says-community-abandoned-by-newsom/509-a2a2f3d6-c510-48e9-af7d-d2e0e682f3f0

V Appendix

Example 1:



Example 2:


ALEJESE
DEL AGUA
AGUA CONTAMINADA
CON AGUAS RESIDUALES
ESTA AGUA PUEDE
CAUSAR ENFERMEDAD

Example 3:



Example 4:

California Coastal Commission Votes To Support State of Emergency Declaration

The California Coastal Commission voted unanimously to send a letter to the Biden Administration requesting a federal State of Emergency declaration to address the escalating environmental and public health crisis caused by ongoing pollution in the Tijuana River. The decisive vote follows declaration requests made in recent weeks by the San Diego City Council, the California Legislature, and San Diego County's congressional delegation.

For decades, pollution from sewage, toxic waste, and debris flowing into the Tijuana River has severely impacted the southern San Diego coastline resulting in beach closures, health risks for communities, and harmful effects on marine ecosystems.

Coastal Commission Chair Caryl Hart credited Mayor Aguirre for her leadership and emphasized the importance of swift action during her remarks stating, "I want to align myself with everything that's been said including the incredible gratitude for everything that Commissioner Aguirre has done at her own personal expense." Chair Hart continued, "Everyone in Imperial Beach is feeling the impacts of this situation. It should be all-hands-on-deck on this very, very serious situation. It's a crisis – an absolute crisis."

Imperial Beach Mayor Paloma Aguirre, who also serves as the San Diego Coast Representative on the California Coastal Commission, led discussion on the item at the Coastal Commission meeting which was held at the Catamaran Hotel in Pacific Beach.

"It is the biggest coastal issue in our entire state," said Mayor Aguirre during her remarks at the commission meeting. "I want to appreciate my colleagues' support and patience... We cannot be complacent because we may think that we're powerless, but I would argue we are powerful."

Recently, Mayor Aguirre sent a letter to White House Chief of Staff Jeff Zients detailing a renewed request for President Biden to declare a federal State of Emergency. The unchecked, ongoing contamination of the Tijuana River continues to negatively impact the health and safety of south San Diego County residents.

"I support the commission sending the strongest language possible that moves us to action on this crisis," said Alternate Commissioner and San Diego resident Gretchen Newsom. "As a sister of San Diego County on the Coastal Commission, this environmental disaster is adversely affecting hard-working families, including as many as 3,000 union members who live in Imperial Beach, and thousands more throughout San Diego's South bay communities. Every resident in California deserves access to clean air, clean water, and pollution-free recreation areas. This crisis has reached a tipping point."

The commission's letter to the Biden Administration will highlight perpetual environmen-

Coastal, Continued on Page 9.

AT HOME **REALTY**

PROPERTY MANAGEMENT
VACATION RENTALS & HOME CONCIERGE
SALES
Nancy Parrett - Broker - Owner
619-368-1898
CA BRE #01256239

Coastal, Continued from Page 4.

tal issues exacerbating the crisis including extreme weather events like heavy rains and severe flooding from earlier this year. The letter will advocate and request a call for immediate, comprehensive solutions in addressing the ongoing environmental crisis.

The Tijuana River has been discharging up to 80 million gallons per day of untreated sewage, industrial waste, and urban runoff, exposing Imperial Beach residents and neighboring communities to high levels of toxic chemicals, pathogens, and harmful gases. Researchers from University of California San Diego and San Diego State University have identified concerning levels of hydrogen sulfide (H_2S) that exceeded state standards during nighttime hours. Combined with other volatile organic compounds, the mixture of polluted air and ocean water has been causing a steady increase in respiratory and gastrointestinal ailments and illnesses in local residents.

The California Coastal Commission is a state agency within the California Natural Resources Agency with quasi-judicial control of protecting and enhancing coastal waters for present and future generations through planning and regulation of development, natural resource protection, and public access along the state's 1,100-mile coastline.

Video from the October 10th Coastal Commission meeting can be found online (Video: California Coastal Commission- Oct. 10, 2024, 9 a.m. CAL-SPAN). The informational update and commission's action on trans-boundary pollution in the Tijuana River Valley begins at timecode 01:12:44.

For further information, please contact: Ryan Trabuco at rtrabuco@imperialbeach.ca.gov or 619.904.1267

Example 5:

The Mayor's Corner

Coastal Commission Joins The Fight For A State Of Emergency On The Tijuana River Crisis

As Mayor of Imperial Beach, I have the privilege of representing our beautiful community on the California Coastal Commission, where I'm also a voice for the entire San Diego coastline. In recent weeks, we've made tremendous strides in our fight against the pollution that has plagued the Tijuana River Valley for decades. The California Coastal Commission, through my leadership and the support of its Chair, Caryl Hart, voted unanimously to push for a federal State of Emergency declaration. This is a monumental step toward securing the urgent resources we need to protect Imperial Beach and our neighboring communities.

The pollution crisis in the Tijuana River has been devastating, with untreated sewage, toxic chemicals, and industrial waste flowing into our coastal waters, threatening the health of our residents and the environment. Our beaches have been closed repeatedly, our air has been filled with concerning levels of hydrogen sulfide, and our community has been left grappling with the severe health impacts of this unchecked pollution.

We cannot wait any longer for federal action, and that is why, at the Coastal Commission's most recent meeting, I urged the commission to join us in calling on the Biden Administration to declare a State of Emergency. Chair Caryl Hart echoed my concerns, calling this situation exactly what it is: a crisis. This unified stance from the Coastal Commission is crucial in

Mayor, Continued on Page 5.



Eagle & Times

Dean Eckenroth

Mayor, Continued from Page 5.

amplifying our voices and pressuring the federal government to step in.

We are not alone in this fight. Last month, all 120 California state legislators voted unanimously to approve a resolution supporting the State of Emergency declaration. The resolution, which calls for immediate federal intervention, was backed by every member of the California State Legislature—Democrats and Republicans alike. This level of bipartisan support is a testament to the gravity of the crisis we face.

On top of that, the City of San Diego has passed a similar resolution, with the support of all its council members. All 18 mayors across San Diego County have also been calling for the declaration since last year. Together, we are showing that this is not just a local issue for Imperial Beach—it's a regional emergency that affects the entire Southern California coast. I'm deeply grateful for the leadership of our state legislators, and all my fellow mayors who have stood with us. Their support strengthens our call to Washington to act now.

We are making progress, but we cannot let up. The health of our community, the vitality of our economy, and the future of our coastline depend on the decisions being made in the coming weeks. I will keep you all updated as we continue to push for the declaration of a State of Emergency and the solutions that will finally put an end to this crisis.

Together, we will make sure Imperial Beach's voice is heard.

Paloma Aguirre,
Mayor of Imperial Beach